



Interference and System Capacity

- Source of interference include: another mobile in the same cell, a call in progress in a neighboring cell, other base station, any non cellular systems.
- The two major types of system-generated cellular interference are co-channel interference and adjacent channel interference.

9

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Co-channel Interference and System Capacity

- Frequency reuse implies that in a given coverage area there are several cells that use the same set of frequencies. These cells are **called co-channel cells**.
- The interference between signals from these cells is called **co-channel interference**.
- To reduce co-channel interference, co-channel cells must be physically separated by a minimum distance.

10

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- The co-channel interference ratio is a function of the radius of the cell (R) and the distance between centers of the nearest co-channel cell (D).

- D/R increased, co-channel interference decreased.
- The co-channel reuse ratio is related to the cluster size. For a hexagonal geometry

$$Q = \frac{D}{R} = \sqrt{3N}$$

11

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- $Q \downarrow$, capacity $\uparrow$
- $Q \downarrow$, co-channel interference $\uparrow$
A trade-off must be made between these two objectives in actual cellular design.
- The signal-to-interference ratio (S/I or SIR) for a mobile receiver can be expressed as

$$\frac{S}{I} = \frac{S}{\sum_{i=1}^{i_0} I_i}$$

- where S is the desired signal power from the desired base station
- I_i is the interference power caused by the i^{th} interfering co-channel cell base station.

12

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The average received power P_r at a distance d is from the transmitting antenna is

$$P_r = P_0 \left(\frac{d}{d_0} \right)^{-n}$$

Or

$$P_r(\text{dBm}) = P_0(\text{dBm}) - 10n \log \left(\frac{d}{d_0} \right)$$

Where P_0 is the power received at a close-in reference point in the far field region of the antenna at a small distance d_0 from the transmitting antenna and n is the path loss exponent.

13

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$$\frac{S}{I} = \frac{R^{-n}}{\sum_{i=1}^{i_0} (D_i)^{-n}}$$

if all the interfering base stations are equidistant from the desired base station and if this distance is equal to the distance D between cell centers

$$\frac{S}{I} = \frac{(D/R)^n}{i_0} = \frac{(\sqrt{3}N)^n}{i_0}$$

relates S/I to the cluster size N , which in turn determines the overall capacity of the system

14

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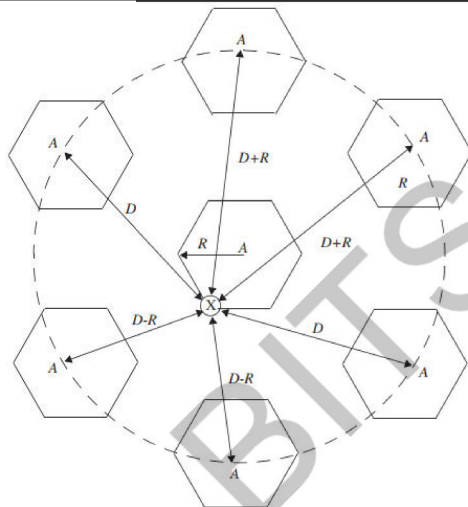


Figure Illustration of the first tier of co-channel cells for a cluster size of $N = 7$. An approximation of the exact geometry is shown here, whereas the exact geometry is given in [Lee86]. When the mobile is at the cell boundary (point X), it experiences worst case co-channel interference on the forward channel. The marked distances between the mobile and different co-channel cells are based on approximations made for easy analysis.

15

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SIR for the worst case can be approximated as

$$\frac{S}{I} = \frac{R^{-4}}{2(D-R)^{-4} + 2(D+R)^{-4} + 2D^{-4}}$$

$$\frac{S}{I} = \frac{1}{2(Q-1)^{-4} + 2(Q+1)^{-4} + 2Q^{-4}}$$

For $N=7$, Q is 4.6, the worst case S/I is 49.56 (17dB) less than 18dB requirement Hence, choose $N = 9$

This entails a significant decrease in capacity. From 1/7 spectrum utilization to 1/9

16

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In practice, a capacity reduction of 7/9 would not be tolerable to accommodate for the worst case situation which rarely occurs.

17



Ex: If a signal-to-interference ratio of 15 dB is required for satisfactory forward channel performance of a cellular system, what is the frequency reuse factor and cluster size that should be used for maximum capacity if the path loss exponent is (a) $n = 4$, (b) $n = 3$? Assume that there are six co-channel cells in the first tier, and all of them are at the same distance from the mobile. Use suitable approximations.

18



(a) $n = 4$

let us consider a seven-cell reuse pattern.

Using Eq. (3.4), co-channel reuse ratio $D/R = 4.583$ Using

Eq. (3.9), signal-to-noise interference ratio

$$S/I = (1/6) \times (4.583)^4 = 75.3 = 18.66 \text{ dB}$$

$N = 7$ can be used.

(b) $n = 3$

$$S/I = (1/6) \times (4.583)^3 = 16.04 = 12.05 \text{ dB}$$

when $N = 12$, $D/R = 6.0$

$$S/I = (1/6) \times (6)^3 = 36 = 15.56 \text{ dB}$$

so, $N = 12$ is used.

19



Channel Planning for Wireless System

- Typically, about 5% of the entire mobile spectrum is devoted to **control channels** (vital for initiating, requesting, or paging a call), while 95% of the spectrum is dedicated to **voice channel** (carrying revenue – generating calls).
- Since control channels are vital, the frequency reuse strategy for them is generally more conservative than for voice channels.
- In example 3.3, 21-cell reuse for control channels, whereas 7-cell reuse for voice channels.

20



Adjacent Channel Interference

- Resulting from signals which are adjacent in frequency to the desired signal.
- The near-far effect: a nearby transmitter captures the receiver of the subscriber or the base station.
- Adjacent channel interference can be minimized through careful filtering and channel assignments.
- A cell need not be assigned channels which are all adjacent in frequency. Neighboring cells avoid the use of adjacent channels.

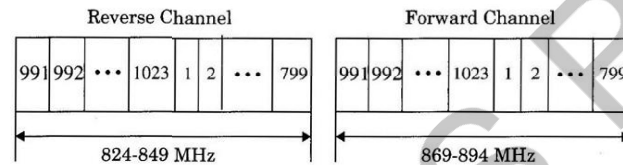
21

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- Ex: This example illustrates how channels are divided into subsets and allocated to different cells so that adjacent channel interference is minimized.
- The United States AMPS system initially operated with 666 duplex channels. In 1989, the FCC allocated an additional 10 MHz (166 channels) of spectrum for cellular services.
- There are now 832 channels used in AMPS.
- The extended band has channels number as 667 through 799, and 990 through 1023.

22

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	Channel Number	Center Frequency (MHz)
Reverse Channel	$1 \leq N \leq 799$	$0.030N + 825.0$
	$991 \leq N \leq 1023$	$0.030(N - 1023) + 825.0$
Forward Channel	$1 \leq N \leq 799$	$0.030N + 870.0$
	$991 \leq N \leq 1023$	$0.030(N - 1023) + 870.0$
(Channels 800-990 are unused)		

Figure Frequency spectrum allocation for the U.S. cellular radio service. Identically labeled channels in the two bands form a forward and reverse channel pair used for duplex communication between the base station and mobile. Note that the forward and reverse channels in each pair are separated by 45 MHz.

24

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- In order to encourage competition, the FCC licensed the channels to two competing operators in every service area.
- The channels used by the two operators are distinguished as *block A* and *block B* channels.
 - Block B is operated by companies which have traditionally provided telephone services (called *wireline operators*).
 - Block A is operated by companies that have not traditionally provided telephone services (called *nonwireline operators*).
- Out of the 416 channels used by each operator, 395 are voice channels and the remaining 21 are control channels.
 - Block A – voice channels 1~312, 667~716, 991~1023.
control channels 313~333.
 - Block B – voice channels 355~666, 717~799.
control channels 334~354.

24

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- One control channel is provided for each bank of trunked voice channels.
- For an AMPS system, a single cell is assigned a single control channel.
- Thus, control channels obey a 21-cell reuse scheme and are assigned over three clusters before being reused, even through the voice channels may be assigned using a seven-cell reuse scheme.

25

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Table AMPS Channel Allocation for A and B Side Carriers

1A	2A	3A	4A	5A	6A	7A	8A	9A	10A	11A	12A	13A	14A	15A	16A	17A	18A	19A	20A	21A	22A	23A	24A	25A	26A	27A	28A	29A	30A	31A	32A	33A	34A	35A	36A	37A	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	
22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42																	
43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63																	
64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84																	
85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105																	
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126																	
127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147																	
148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	
169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	
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313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350
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712	713	714	715	716	717	-	-	-	-	-	991	992	993	994	995	996	997	998	999	1000	1001	1002	1003	1004	1005	1006	1007	1008	1009	1010	1011	1012	1013	1014	1015	1016	1017
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27

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- Each of the 395 voice channels are divided into 21 subsets, each containing about 19 channels.
- In each subset, the closest adjacent channel is 21 channels away.
- In a seven-cell reuse system, each cell uses three subsets of channels.
- This channel assignment scheme is illustrated in Table 3.2.
- The total number of voice channels in a cell is about 57.
- The shaded set of numbers correspond to the control channels.

26

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Power Control for Reducing Interference



- The power levels transmitted by every subscriber unit are under constant control by the serving base stations.
- To ensure that each mobile transmits the smallest power necessary to maintain a good quality link on the reverse channel.
- Prolong battery life and reduce the reverse channel S/I .

28

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Trunking and Grade of Service

- The concept of **trunking** allows a large number of user to share the relatively small number of channels in a cell by providing access to each user, on demand, from a pool of available channels.
- A user is **blocked**, i.e., denied access to the system.
- In some system, a queue may be used to hold the requesting users until a channel becomes available.

29

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The **grade of service (GOS)** is a measure of the ability of a user to access a trunked system during the busiest hour, and is typically given as the likelihood that a call is blocked, or the likelihood of a call experiencing a delay greater than a certain queuing time.

31

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- One **Erlang** represents the amount of traffic intensity carried by a channel that is completely occupied (i.e. one call-hour per hour or one call-minute per minute).

For example:

0.5 Erlangs of traffic – a radio channel that is occupied for 30 minutes during an hour.

30

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Definitions of Common Terms Used in Trunking Theory

Set-up Time – The time required to allocate a trunked radio channel to a requesting user.

Blocked Call – Call which cannot be completed at time of request, due to congestion. Also referred to as a *lost call*.

Holding Time – Average duration of a typical call. Denoted by H (in seconds).

Traffic Intensity – Measure of channel time utilization, which is the average channel occupancy measured in Erlangs. This is a dimensionless quantity and may be used to measure the time utilization of single or multiple channels. Denoted by A .

32

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Load – Traffic intensity across the entire trunked radio system, measured in Erelang.

Grade of Service (GOS) – A measure of congestion which is specified as the probability of a call being blocked (for Erlang B), or the probability of a call being delayed beyond a certain amount of time (for Erlang C).

Request Rate – The average number of call requests per unit time. Denoted by λ seconds⁻¹

33



- Traffic intensity A_u of each user is given by

$$A_u = \lambda H$$

- Total traffic intensity A for U users

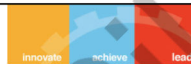
$$A = UA_u$$

- Traffic intensity per channel A_c

$$A_c = UA_u / C$$

- The maximum possible carried traffic is the total number of channels, C , in Erlangs.

34



There are two types of trunked systems.

1. Blocked calls cleared: This is known as an M/M/m/m queue and leads to the derivation of the Erlang B formula.

$$P_r[\text{blocking}] = \frac{\frac{A^C}{C!}}{\sum_{k=0}^C \frac{A^k}{k!}} = GOS$$

where C is the number of trunked channels offered by a trunked radio system and A is the total offered traffic

35



Table Capacity of an Erlang B System

Number of Channels C	Capacity (Erlangs) for GOS			
	= 0.01	= 0.005	= 0.002	= 0.001
2	0.153	0.105	0.065	0.046
4	0.869	0.701	0.535	0.439
5	1.36	1.13	0.900	0.762
10	4.46	3.96	3.43	3.09
20	12.0	11.1	10.1	9.41
24	15.3	14.2	13.0	12.2
40	29.0	27.3	25.7	24.5
70	56.1	53.7	51.0	49.2
100	84.1	80.9	77.4	75.2

36





2. Blocked calls delayed

Erlang C formula

$$P_r[\text{delay} > 0] = \frac{A^C}{A^C + C! \left(1 - \frac{A}{C}\right) \sum_{k=0}^{C-1} \frac{A^k}{k!}}$$

37



- The GOS of a trunked system is given by

$$P_r[\text{delay} > t] = P_r[\text{delay} > 0] P_r[\text{delay} > t | \text{delay} > 0] \\ = P_r[\text{delay} > 0] \exp(-(C - A)t/H)$$

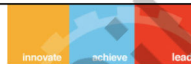
- The average delay D for all calls is

$$D = P_r[\text{delay} > 0] \frac{H}{C - A}$$

where the average delay for those calls which are queued is given by $H/(C - A)$.

38

Improving Coverage and Capacity in Cellular Systems



Cell Splitting

- Cell splitting is the process of subdividing a congested cell into smaller cells.
- Since it increases the number of times that channels are reused (or decreases the cluster area), cell splitting increases the capacity of a cellular system.
- Cell splitting generally does not upset the channel allocation scheme required to maintain the minimum co-channel reuse ratio Q between co-channel cells.

39

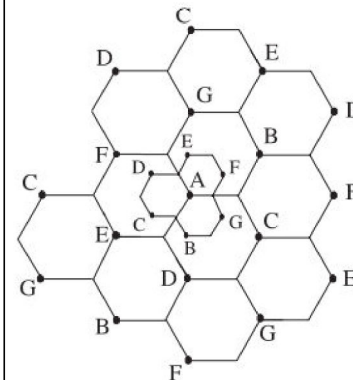


Figure Illustration of cell splitting.

$$P_r[\text{at old cell boundary}] \propto P_{t1} R^{-n} \\ = P_r[\text{at old cell boundary}] \propto P_{t2} (R/2)^{-n}$$

$$P_{t2} = \frac{P_{t1}}{16}$$

The transmit power must be reduced by 12 dB in order to fill in the original coverage area with microcells, while maintaining the S/I requirement.

40



Sectoring

- By decreasing the cell radius R and keeping the co-channel reuse ratio D/R unchanged, cell splitting increases the number of channels per unit area.
- Another way to increase capacity is to keep the cell radius unchanged and seek methods to decrease the D/R ratio.
- Sectoring increases SIR so that the cluster size may be reduced.

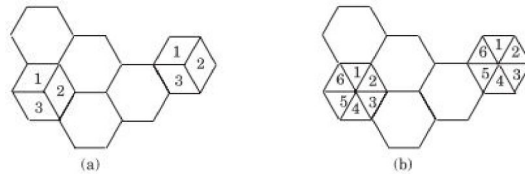


Figure (a) 120° sectoring; (b) 60° sectoring.

41

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- The penalty from sectoring:
 - an increased number of antennas at each base station.
 - a decrease in trunking efficiency due to the channel sectoring.
 - the increased number of handoffs – usually is not a major concern.
- Some operators shy away from sectoring approach, because it breaks up the available trunked pool into several small pools, and decreases trunking efficiency.

43

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S/I increases to 24.2 dB.

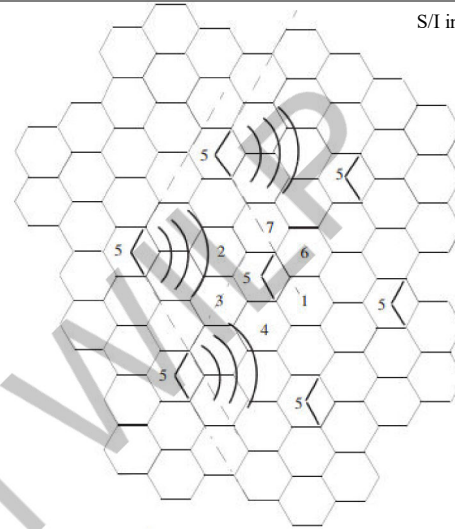


Figure : Illustration of how 120° sectoring reduces interference from co-channel cells. Out of the 6 co-channel cells in the first tier, only two of them interfere with the center cell. If omnidirectional antennas were used at each base station, all six co-channel cells would interfere with the center cell.

42

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Repeaters for Range Extension

- Repeaters – to provide dedicated coverage for hard-to-reach areas, such as within buildings, or in valleys or tunnels.

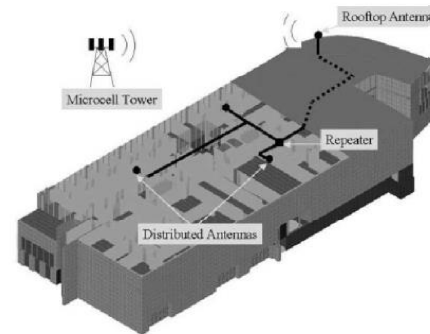


Figure Illustration of how a distributed antenna system (DAS) may be used inside a building. Figure produced in SitePlanner®. (Courtesy of Wireless Valley Communications Inc.)

44

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Microcell Zone Concept

- When sectoring is employed, the increased number of handoffs results in an increased load in MSC. A solution to this problem is a Microcell Zone Concept.

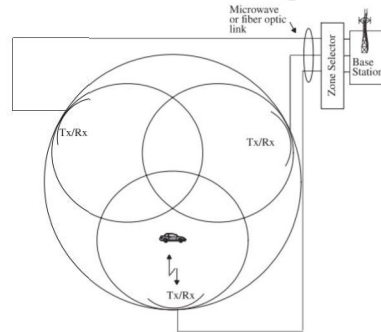


Figure The microcell concept (adapted from [Lee91b] © IEEE).

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45

- Antennas are placed at the outer edges of the cell. Each of the three zone sites are connected to a base station. Any base station channel may be assigned to any zone.
- As a mobile travels from one zone to another within the cell, it retains the same channel. The base station simply switches the channel to a different zone site.
- The advantage of the zone cell technique is that the co-channel interference in the cellular system is reduced

$$S/I \geq 18 \text{ dB} \Rightarrow N = 7 \Rightarrow D/R = 4.6$$

46

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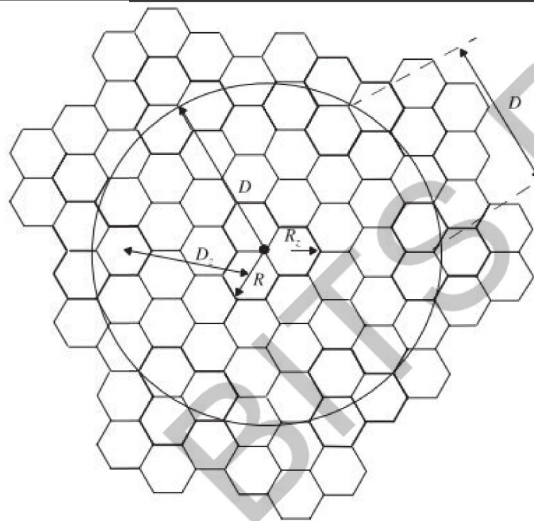


Figure Define D , D_z , R , and R_z for a microcell architecture with $N = 7$. The smaller hexagons form zones and three hexagons (outlined in bold) together form a cell. Six nearest co-channel cells are shown.

$$D_z / R_z = 4.6$$

From the geometry

$$D/R = 3 \Rightarrow N = 3$$

$$7/3 = 2.33 \quad \text{Times increase in capacity}$$

No loss in trunking efficiency is experienced.

Many communication system use this technique.

48

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HOME WORK 2.1

T1: Problems: 3.8, 3.10, 3.12

49

IMP Note to Self



IMP Note to Self



IMP Note to Students

- It is important to know that just login to the session does not guarantee the attendance.
- Once you join the session, continue till the end to consider you as present in the class.
- IMPORTANTLY, you need to make the class more interactive by responding to Professors queries in the session.
- **Whenever Professor calls your number / name ,you need to respond, otherwise it will be considered as ABSENT**




Wireless & Mobile Communication
(CSI ZG520)

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Modular Content Structure

- M1: Introduction to Wireless Systems and Modern Wireless Communication System.
- M2: The Cellular Concept- System Design Fundamentals
- M3: Cellular Network
- M4: Mobile Radio Propagation: Large scale path Loss_ Part 1**
- M5: Mobile Radio Propagation: Large scale path Loss_ Part 2
- M6: Mobile Radio Propagation: Large scale path Loss_ Part 3
- M7: Small Scale fading_ Part 1
- M8: Small Scale fading_ Part 1

Modular Content Structure

- M9: Modulation Techniques for Mobile Radio_ Amplitude Modulation
- M10: Modulation Techniques for Mobile Radio_ Angle Modulation
- M11: Digital Modulation Techniques_1
- M12: Digital Modulation Techniques_2
- M13: Equalization & Diversity Techniques
- M14: Concepts of OFDM & MIMO
- M15: Multiple Access Techniques
- M16: Wireless Networking

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5

Text Book:

- T1: “Wireless Communication Principles and Practice” by Theodore. S. Rappaport Second Ed., Prentice Hall of India, 2008.
- T2: “Wireless Communications and Networks” by William Stallings – Pearson Education Ltd.
- R1: “Mobile Communication” by Jochen H. Schiller, Addison –Wesley, Pearson Education Ltd., 2000.
- R2: “Wireless and Mobile Network Architectures” by Yi-Bing Lin and Imrich Chaltamac, John Wiley and Sons, 2001.

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6



Mobile Radio Propagation: Large Scale Path Loss

7

M4: Mobile Radio Propagation: Large scale path Loss

- | Type | Description/Plan/Reference |
|----------|--|
| RL 3.1.1 | Mobile Radio Propagation |
| RL_3.1.2 | Propagation Mechanism of Radio waves |
| RL_3.2.1 | Free space Propagation model and Two Ray ground reflected model |
| CS 4 | A brief summary of recorded lectures, Reflection co-efficient for parallel and Horizontal polarization of radio waves, Problems on Brewster angle, Radar cross section model. Problems on Free space propagation model |
| HW 4 T1: | Problems: 4.1, 4.2, 4.4,4.5,4.6 |
| LE 4 | MATLAB experiment based on the CS 4.0 |

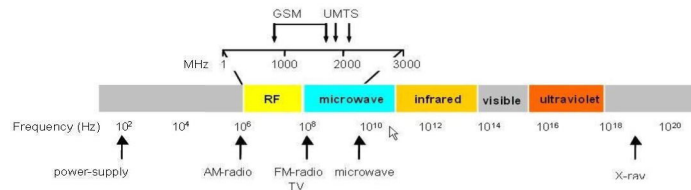
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8



Wave Propagation

- Radio, microwave, infrared and visible light portions of the spectrum can all be used to transmit information
 - By modulating the amplitude, frequency, or phase of the waves.
- The amount of information a wireless channel can carry is related to its bandwidth
- Wavelength dictates the optimum size of the receiving antenna



9

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Introduction to Radio Wave Propagation

- The mobile radio channel places fundamental limitations on the performance of wireless communication systems.
- Radio channels are extremely random and do not offer easy analysis.
- Modeling radio channel is important for:
 - Determining the coverage area of a transmitter
 - Determine the transmitter power requirement
 - Determine the battery lifetime
 - Finding modulation and coding schemes to improve the channel quality
 - Determine the maximum channel capacity

10

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Introduction to Radio Wave Propagation

- The mechanisms behind electromagnetic wave propagation are diverse, but can generally be attributed to **reflection**, **diffraction**, and **scattering**.
- Propagation models have traditionally focused on predicting the **average received signal strength** at a given distance from the transmitter, as well as the **variability of the signal strength** in close spatial proximity to a particular location.

11

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Introduction to Radio Wave Propagation

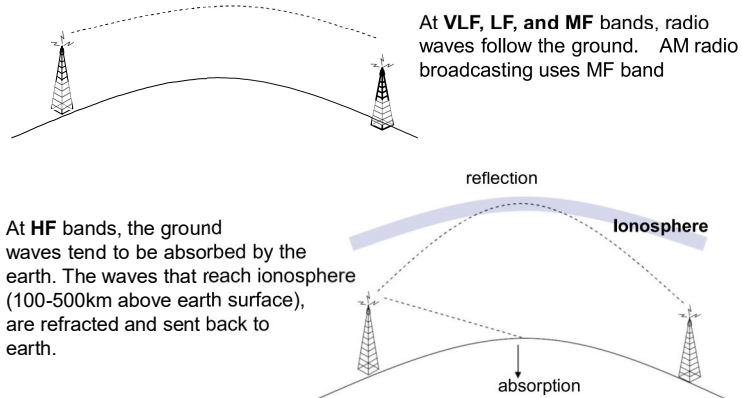
- Propagation models that predict the mean signal strength for an arbitrary transmitter-receiver (T-R) separation distance are useful in estimating the radio coverage area of a transmitter and are called **large-scale propagation models**.
- On the other hand, propagation models that characterize the rapid fluctuations of the received signal strength over very short travel distances (a few wavelengths) or short time durations (on the order of seconds) are called **small-scale or fading models**.

12

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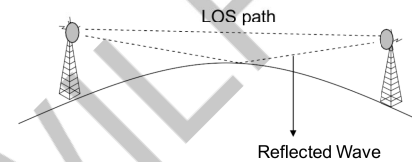
Basics - Propagation



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Basics - Propagation

VHF Transmission



- Directional antennas are used
- Waves follow more direct paths
- LOS: Line-of-Sight Communication
- Reflected wave interfere with the original signal

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Basics - Propagation

- Waves behave more like light at higher frequencies
 - Difficulty in passing obstacles
 - More direct paths
- They behave more like radio at lower frequencies
 - Can pass obstacles

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Radio Propagation Models

- Transmission path between sender and receiver could be
 - Line-of-Sight (LOS)
 - Obstructed by buildings, mountains and foliage
- Even speed of motion effects the fading characteristics of the channel

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Free-Space Propagation Model



- Predict the received signal strength when transmitter and receiver have clear, unobstructed LOS path between them.
 - Ex: Satellite communication system, microwave LOS system
- The received power decays as a function of T-R separation raised to some power.
- Free space power received by a receiver antenna is given by **Friis free-space equation**

$$P_r(d) = (P_t G_t G_r \lambda^2) / ((4\pi)^2 d^2 L)$$

17

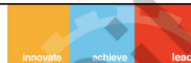
Free-Space Propagation Model (cont.)



- P_t is transmitted power
- $P_r(d)$ is the received power
- G_t is the transmitter antenna gain (dimensionless quantity)
- G_r is the receiver antenna gain (dimensionless quantity)
- d is T-R separation distance in meters
- L is system loss factor not related to propagation ($L \geq 1$)
 - $L = 1$ indicates no loss in system hardware (for our purposes we will take $L = 1$, so we will ignore it in our calculations).
- λ is wavelength in meters.

18

Free-Space Propagation Model (cont.)



- The gain of an antenna G is related to its effective aperture A_e by $G = 4\pi A_e / \lambda^2$ where
 - A_e is related to the physical size of the antenna
 - λ is related to the carrier frequency ($\lambda = c/f = 2\pi c / \omega_c$) where
 - f is carrier frequency in Hertz
 - c is speed of light in meters/sec
 - ω_c is carrier frequency in radians per second
- **Isotropic** radiator generally considered an reference antenna in wireless systems; radiates power with unit gain uniformly in all directions.
- **Effective isotropic radiated power (EIRP)** is the amount of power that a theoretical isotropic antenna emits to produce peak power density in the direction of maximum antenna gain.

$$EIRP = P_t G_t$$
- Antenna gains are given in units of **dBi** (dB gain with respect to an isotropic antenna)

19

What is Decibel (dB)



- A logarithmic unit used to describe a ratio between two values of a physical quantity (usually measured in units of power or intensity)
- The ratio of two values P_1 and P_2 in dB is
 - ✓ $10 \log (P_1/P_2)$ dB
- Example: $P_1 = 100 \text{ W}$ and $P_2 = 1 \text{ W}$

$$\text{The ratio is } 10 \log(100/1) = \mathbf{20 \text{ dB}}$$
- dB unit is generally used to describe ratios of numbers with modest size.

20





dBm

- Indicates power ratio in dB with 1mW as the reference power
- Example: Transmit power = 100W in dBm is

$$\begin{aligned}\text{Transmit power (dBm)} &= 10 \log(100 \text{ W} / 1 \text{ mW}) \\ &= 10 \log(100,000) \\ &= 50 \text{ dBm}\end{aligned}$$

Similarly

1 mW = 0 dBm	1 W = 30 dBm	10 W = 40 dBm
100 W = 50 dBm	$10^6 \text{ W} = 90 \text{ dBm}$	

21

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dBW

- Indicates power ratio in dB with 1 W as the reference power level.
- Example: Transmit power = 100 W in dBW is

$$\begin{aligned}\text{Transmit power (dBW)} &= 10 \log(100 \text{ W} / 1 \text{ W}) \\ &= 10 \log 100 \\ &= 20 \text{ dBW}\end{aligned}$$

Similarly

1 mW = -30 dBm	1 W = 0 dBm	10 W = 10 dBm
100 W = 20 dBm	$10^6 \text{ W} = 60 \text{ dBm}$	

22

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Free-Space Path Loss

- Path loss is defined as the difference (in dB) between the effective transmitted power and the received power
- Free-space path loss is defined as the path loss of the free-space model

$$PL(\text{dB}) = 10 \log(P_t/P_r) = -10 \log[(G_t G_r \lambda^2)/(4\pi)^2 d^2]$$
- Friis equation holds when distance d is in the far-field of the transmitting antenna
- The **far-field** or **Fraunhofer region** of a transmitting antenna is defined as the region beyond the **far-field distance** d_f given by:
 - $d_f = 2D^2/\lambda$, D is the largest physical dimension of the antenna
 - Additionally $d_f \gg D$ and $d_f \gg \lambda$

23

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Reference Distance, d_0

- Friis free space eq. does not hold for $d = 0$
- Received power reference point, d_0 is used

$$d_f \leq d_0 \leq d$$
- d_0 should be smaller than any practical distance a mobile system uses
- The power received in free space at a distance greater than d_0 is

$$P_r(d) = P_r(d_0)(d_0/d)^n \quad \text{where } d_f \leq d_0 \leq d$$
- Reference distance d_0 for practical systems:
 - For frequencies in the range 1 to 2 GHz
 - 1 m in indoor environments
 - 100 m to 1 km in outdoor environments

24

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- Ex: A transmitter produces 50W of power.
 - A) Express the transmit power in dBm
 - B) Express the transmit power in dBW
 - C) If d_0 is 100m and the received power at that distance is 0.0035mW, then find the received power level at a distance of 10km.
 - Assume that the transmit and receive antennas have unity gains.

25

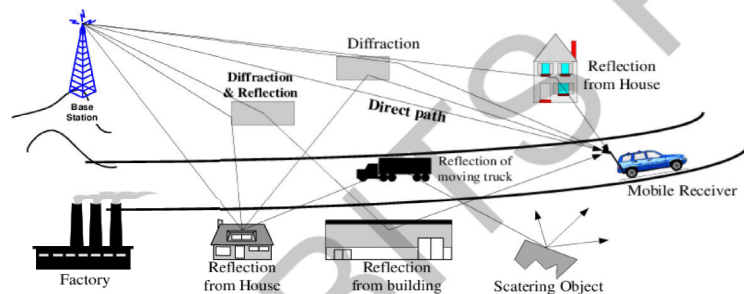
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- Sol
- A)
 - $P_t(W)$ is 50W.
 - $P_t(dBm) = 10\log[P_t(mW)/1mW]$ $P_t(dBm) = 10\log(50 \times 1000)$ $P_t(dBm) = 47 \text{ dBm}$
- B)
 - $P_t(dBW) = 10\log[P_t(W)/1W]$ $P_t(dBW) = 10\log(50)$ $P_t(dBW) = 17 \text{ dBW}$
- $P_r(d) = P_r(d_0)(d_0/d)^2$
- Substitute the values into the equation:
 - $P_r(10km) = P_r(100m)(100m/10km)^2$ $P_r(10km) = 0.0035mW(10^{-4})$
 $P_r(10km) = 3.5 \times 10^{-10}W$
 - $P_r(10km) [dBm] = 10\log(3.5 \times 10^{-10}W/1mW)$
 $= 10\log(3.5 \times 10^{-7})$
 $= -64.5dBm$

26

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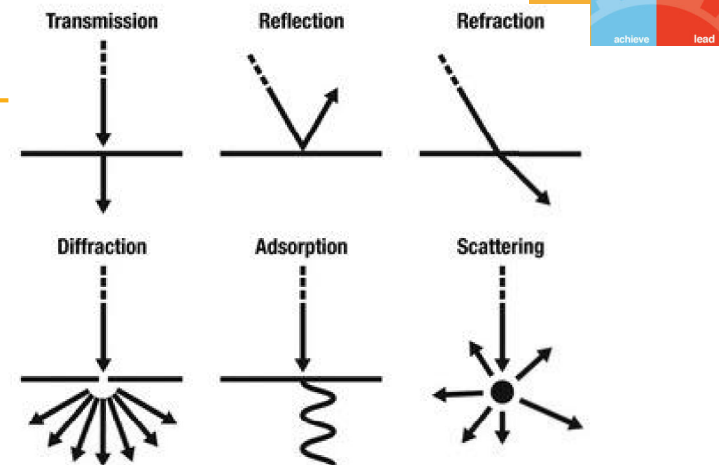
Radio propagation mechanisms



Source: Radio Frequency and Wireless Communications - Scientific Figure on Research Gate

27

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Source: <https://physicsweekly.weebly.com/uploads/2/5/8/4/25849299/6320607.jpg?332>

28

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Propagation Mechanism

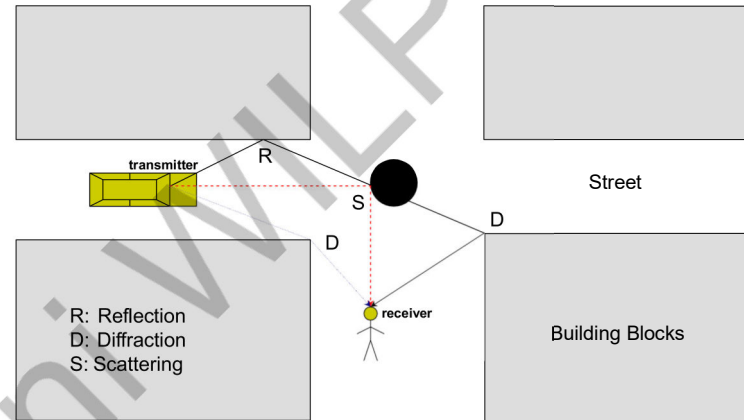
Reflection, Diffraction and Scattering

- **Reflection** occurs when a propagating electromagnetic wave impinges upon an object which has **very large dimensions** when compared to the wavelength of the propagating wave.
- **Diffraction** occurs when the radio path between the transmitter and receiver is obstructed by a surface that has sharp irregularities (edges).
- **Scattering** occurs when the medium through which the wave travels consists of objects with dimensions that are small compared to the wavelength, and where the number of obstacles per unit volume is large.

29

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Three Radio Propagation Mechanisms



30

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Reflection

Fresnel Reflection Coefficient (Γ)

It gives the relationship between the electric field intensity of the reflected and transmitted waves to the incident wave in the medium of origin.

The Reflection Coefficient is a function of the material properties

It depends on

Wave Polarization

Angle of Incidence

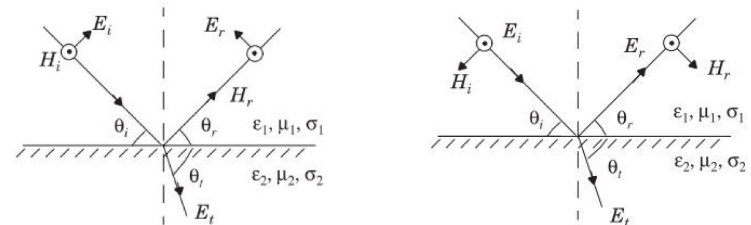
Frequency of the propagating wave

31

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Reflection from Dielectric

The behavior for arbitrary directions of polarization can be studied by considering the two distinct cases.



(a) E-field in the plane of incidence

(b) E-field normal to the plane of incidence

Figure Geometry for calculating the reflection coefficients between two dielectrics.

32

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- The behavior for arbitrary directions of polarization is illustrated through the two distinct cases in the figure
- Case 1:** The E - field polarization is **parallel** with the plane of Incidence i.e. the E - field has a vertical polarization, or normal component with respect to the reflecting surface
- Case 2:** The E - field polarization is **perpendicular** to the plane of incidence i.e. the E - field is parallel to the reflecting surface (normal to the page and pointing out of it towards the reader)

33

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Table Material Parameters at Various Frequencies

Material	Relative Permittivity ϵ_r	Conductivity σ (s/m)	Frequency (MHz)
Poor Ground	4	0.001	100
Typical Ground	15	0.005	100
Good Ground	25	0.02	100
Sea Water	81	5.0	100
Fresh Water	81	0.001	100
Brick	4.44	0.001	4000
Limestone	7.51	0.028	4000
Glass, Corning 707	4	0.0000018	1
Glass, Corning 707	4	0.000027	100
Glass, Corning 707	4	0.005	10000



- The dielectric constant ϵ of a perfect (lossless) dielectric is given by

$$\epsilon = \epsilon_0 \epsilon_r$$

where ϵ_r is the relative permittivity

and $\epsilon_0 = 8.85 * 10^{-12}$ F/ m

- The dielectric constant ϵ for a power absorbing, lossy dielectric is

$$\epsilon = \epsilon_0 \epsilon_r - j \epsilon''$$

where $\epsilon'' = \sigma / 2\pi f$

34

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- In the case when the first medium is free space and $\mu_1 = \mu_2$ the Reflection coefficients for the two cases of vertical and horizontal polarization can be simplified to

$$\Gamma_{\parallel} = \frac{-\epsilon_r \sin \theta_i + \sqrt{\epsilon_r - \cos^2 \theta_i}}{\epsilon_r \sin \theta_i + \sqrt{\epsilon_r - \cos^2 \theta_i}}$$

$$\Gamma_{\perp} = \frac{\sin \theta_i - \sqrt{\epsilon_r - \cos^2 \theta_i}}{\sin \theta_i + \sqrt{\epsilon_r - \cos^2 \theta_i}}$$

36

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Brewster Angle

- It is the angle at which no reflection occurs in the medium of origin
- It occurs when the incident angle θ_B is such that the Reflection Coefficient $\Gamma_{||} = 0$

$$\sin(\theta_B) = \sqrt{\frac{\epsilon_1}{\epsilon_1 + \epsilon_2}}$$

- For the case when the first medium is free space and the second medium has a relative permittivity ϵ_r , the above equation can be expressed as (i.e., $\epsilon_1 = \epsilon_0$, $\epsilon_2 = \epsilon_0 \epsilon_r$)

$$\sin(\theta_B) = \sqrt{\frac{\epsilon_r - 1}{\epsilon_r^2 - 1}}$$

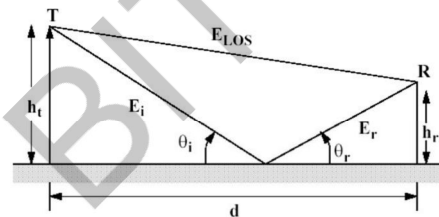
37

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Ground Reflection Model (cont.)

- The total received E-field, E_{TOT} is a result of the direct LOS component E_{LOS} and the ground reflected component E_r

$$E_{TOT} = E_{LOS} + E_r$$



39

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Ground Reflection (Two-Ray) Model

In a mobile radio channel, a single direct path between the BS and a mobile is seldom the only physical means for propagation and the Free space propagation model is inaccurate in most cases when used alone. Two-ray model is

- Based on geometric optics and it considers both the direct path and a ground reflected propagation path
- Reasonably accurate for predicting the large scale signal strength over distances of several kilometers for mobile radio systems that use tall towers.

38

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Ground Reflection (Two-Ray) Model

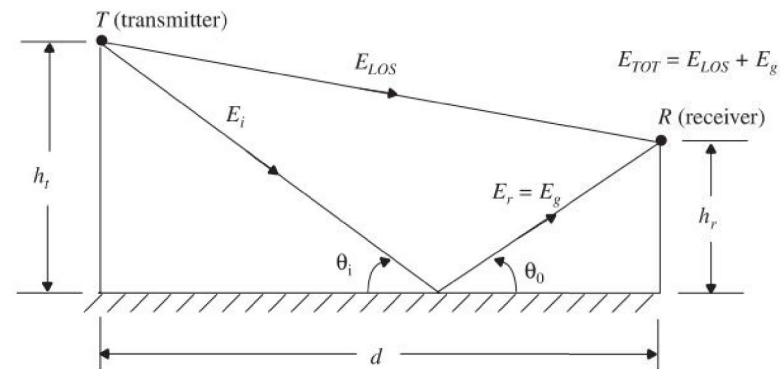


Figure Two-ray ground reflection model.

40

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- $E_{TOT} = E_{LOS} + E_g$
- If E_0 is the free space E-field at a reference distance d_0 from the transmitter, then for $d > d_0$,

$$E(d, t) = \frac{E_0 d_0}{d} \cos(\omega_c(t - \frac{d}{c})) \quad (d > d_0)$$

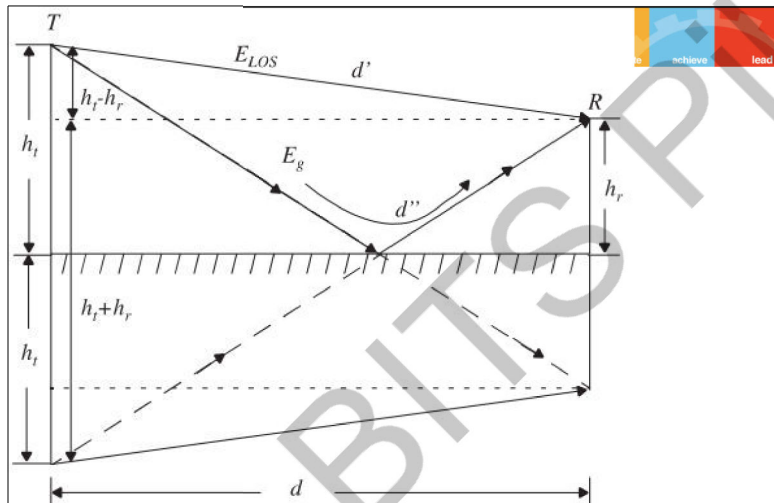
⇒

$$E_{LOS}(d', t) = \frac{E_0 d_0}{d'} \cos(\omega_c(t - \frac{d'}{c}))$$

$$E_g(d'', t) = \Gamma \frac{E_0 d_0}{d''} \cos(\omega_c(t - \frac{d''}{c}))$$

41

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The method of images is used to find the path difference between the line-of-sight and the ground reflected paths.

43

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The laws of reflection give

$$\theta_i = \theta_o$$

$$E_g = \Gamma E_i$$

$$E_i = (1 + \Gamma) E_o$$

For small values of θ_i (i.e., grazing incidence), as shown in

- Assuming perfect horizontal E-field polarization

$$E_{TOT}(d, t) = \frac{E_0 d_0}{d'} \cos(\omega_c(t - \frac{d'}{c})) + (-1) \frac{E_0 d_0}{d''} \cos(\omega_c(t - \frac{d''}{c}))$$

using the method of images in next page,

42

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$$\Delta = d'' - d' = \sqrt{(h_t + h_r)^2 + d^2} - \sqrt{(h_t - h_r)^2 + d^2}$$

$$\approx \frac{2h_t h_r}{d} \quad (\because d \gg (h_t + h_r))$$

$$\theta_\Delta = \frac{2\pi\Delta}{\lambda} = \frac{\Delta\omega_c}{c}$$

$$\tau_d = \frac{\Delta}{c} = \frac{\theta_\Delta}{2\pi f_c}$$

44

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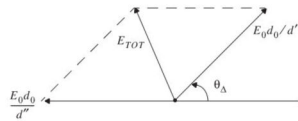
- As d becomes large, the difference between the distances d' and d'' becomes very small

$$\left| \frac{E_0 d_0}{d} \right| \approx \left| \frac{E_0 d_0}{d'} \right| \approx \left| \frac{E_0 d_0}{d''} \right|$$

E_{LOS} and E_g differ only in phase.

- At $t = d''/c$

$$\begin{aligned} E_{TOT}(d, t = \frac{d''}{c}) &= \frac{E_0 d_0}{d'} \cos\left(\omega_c \left(\frac{d'' - d'}{c}\right)\right) - \frac{E_0 d_0}{d''} \cos 0^\circ \\ &= \frac{E_0 d_0}{d'} \angle \theta_\Delta - \frac{E_0 d_0}{d''} \\ &\approx \frac{E_0 d_0}{d} [\angle \theta_\Delta - 1] \end{aligned}$$



45

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- $E_{TOT}(d)$ decays in an oscillatory fashion, with local maxima being 6dB greater than the free space value and the local minima plummeting to $-\infty$ dB (although in reality this never happens).

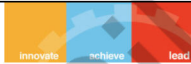
$$|E_{TOT}(d)| = \sqrt{\left(\frac{E_0 d_0}{d}\right)^2 (\cos \theta_\Delta - 1)^2 + \left(\frac{E_0 d_0}{d}\right)^2 \sin^2 \theta_\Delta}$$

$$|E_{TOT}(d)| = \frac{E_0 d_0}{d} \sqrt{2 - 2 \cos \theta_\Delta}$$

$$|E_{TOT}(d)| = 2 \frac{E_0 d_0}{d} \sin\left(\frac{\theta_\Delta}{2}\right)$$

46

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When ever $d > \frac{20\pi h_t h_r}{3\lambda} \approx \frac{20h_t h_r}{\lambda}$

The received E- field can be approximated

$$E_{TOT}(d) \approx \frac{2E_0 d_0}{d} \frac{2\pi h_t h_r}{\lambda d} \approx \frac{k}{d^2} V/m$$

The power received at distance d is given by

$$P_r = P_t G_t G_r \frac{h_t^2 h_r^2}{d^4}$$

For large T- R distances $d \gg \sqrt{h_t h_r}$, so received power falls off to the 4th power of d , or at 40 db/ decade

47

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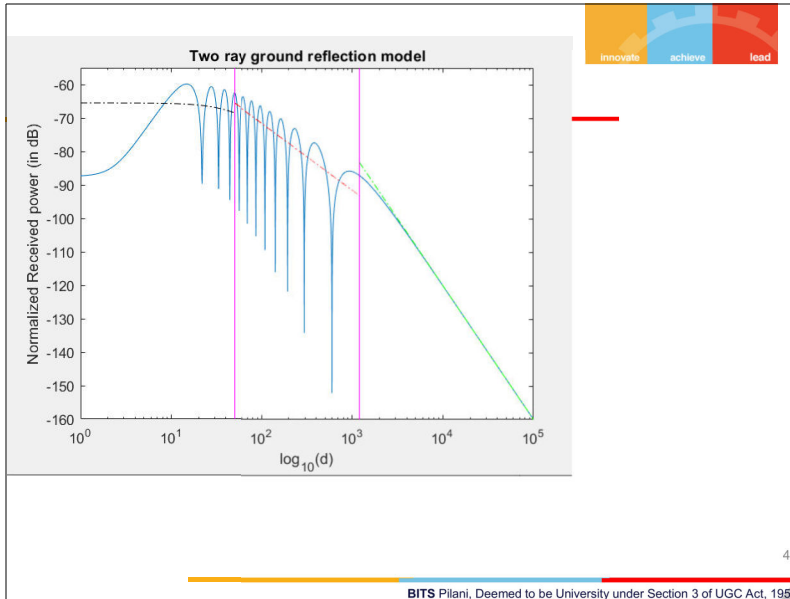
- This power loss is much more than that in free space
- At large values of d , the received power and path loss become independent of frequency.
- The path loss for the 2- ray model in db

$$PL(dB) = 40 \log d - (10 \log G_t + 10 \log G_r + 20 \log h_t + 20 \log h_r)$$

48

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49

Radar cross section model

- The radar cross section (RCS) of a scattering object is defined as the ratio of the power density of the signal scattered in the direction of the receiver to the power density of the radio wave incident upon the scattering object, and has units of square meters.
- Bistatic radar equation**

$$P_R \text{ (dBm)} = P_T \text{ (dBm)} + G_T \text{ (dBi)} + 20\log(\lambda) + \text{RCS} [\text{dBm}^2] \\ - 30\log(4\pi) - 20\log d_T - 20\log d_R$$

50

Practical link budget design using path loss models

Log-distance Path Loss Model

$$\overline{PL}(d) \propto \left(\frac{d}{d_0}\right)^n$$

or

$$\overline{PL}(dB) = \overline{PL}(d_0) + 10n \log\left(\frac{d}{d_0}\right)$$

n is the path loss exponent which indicates the rate at which the path loss increases with distance,

51

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Log-normal Shadowing:

$$PL(d)[dB] = \overline{PL}(d) + X_\sigma = \overline{PL}(d_0) + 10n \log\left(\frac{d}{d_0}\right) + X_\sigma$$

X_σ is the Zero-mean Gaussian distributed random variable with standard deviation σ (also in dB)

$PL(d)$ is a random variable with a normal distribution. Define

$$Q(z) = \frac{1}{\sqrt{2\pi}} \int_z^\infty \exp\left(-\frac{x^2}{2}\right) dx = \frac{1}{2} \left[1 - \text{erf}\left(\frac{z}{\sqrt{2}}\right) \right]$$

The probability that the received signal level will exceed a certain value γ can be calculated from the cumulative density function as

$$\Pr[P_r(d) > \gamma] = Q\left(\frac{\gamma - \overline{P_r}(d)}{\sigma}\right)$$

52

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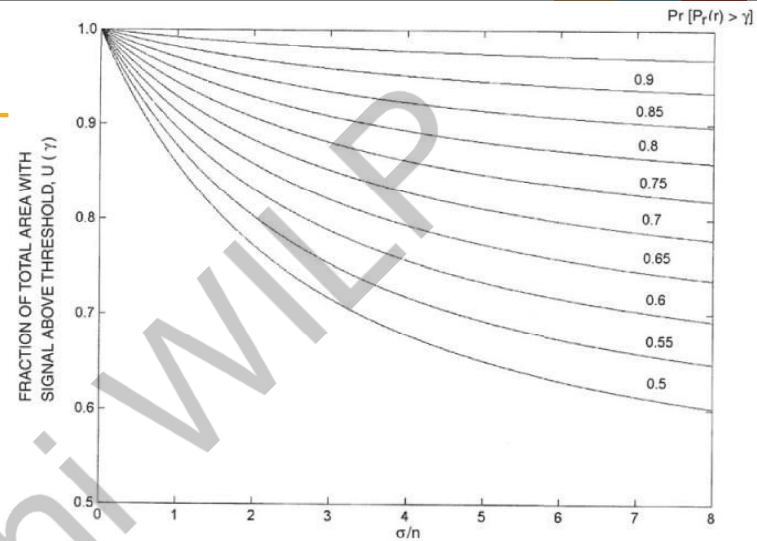
Determination of Percentage of Coverage Area

$$U(\gamma) = \frac{1}{\pi R^2} \int \Pr[P_r(r) > \gamma] dA = \frac{1}{\pi R^2} \int_0^{2\pi R} \int_0^R \Pr[P_r(r) > \gamma] r dr d\theta$$

$$U(\gamma) = \frac{1}{2} \left(1 + \exp\left(\frac{1}{b^2}\right) \left[1 - \operatorname{erf}\left(\frac{1}{b}\right) \right] \right)$$

53

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Family of curves relating fraction of total area with signal above threshold, $U(\gamma)$ as a function of probability of signal above threshold on the cell boundary.

1

3

Problems

Q: Calculate the Brewster angle for a wave impinging on ground having a permittivity of $\epsilon_r = 4$.

$$\sin(\theta_i) = \frac{\sqrt{(4)-1}}{\sqrt{(4)^2-1}} = \frac{\sqrt{3}}{\sqrt{15}} = \frac{1}{\sqrt{5}}$$

$$\theta_i = \sin^{-1} \frac{1}{\sqrt{5}} = 26.56^\circ$$

55

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Q: A mobile is located 5 km away from a base station and uses a vertical $\lambda/4$ monopole antenna with a gain of 2.55 dB to receive cellular radio signals. The E-field at 1 km from the transmitter is measured to be 3 V/m. The carrier frequency used for this system is 900 MHz.

- Find the length and the effective aperture of the receiving antenna.
- Find the received power at the mobile using the two-ray ground reflection model assuming the height of the transmitting antenna is 50 m and the receiving antenna is 1.5 m above ground.

56

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T-R separation distance = 5 km ,
E-field at a distance of 1km = 10^{-3} V/m

Frequency of operation, $f = 900\text{MHz}$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{900 \times 10^6} = 0.333\text{m}$$

$$(a) L = \lambda / 4 = 0.333 / 4 = 0.0833\text{m} = 8.33\text{cm}$$

$$\text{Effective aperture of antenna} = 0.016\text{m}^2$$

$$(b) \text{ since } d \gg \sqrt{h_t h_r}$$

$$E_R(d) \approx \frac{2E_0 d_0}{d} \frac{2\pi h_t h_r}{\lambda d} \approx \frac{k}{d^2} \text{ V/m}$$

$$= \frac{2 \times 10^{-3} \times 1 \times 10^3}{5 \times 10^3} \left[\frac{2\pi(50)(1.5)}{0.333(5 \times 10^3)} \right] = 113.1 \times 10^{-6} \text{ V/m}$$

$$P_r(d) = \frac{(113.1 \times 10^{-6})^2}{377} \left[\frac{1.8(0.333)^2}{4\pi} \right]$$

$$P_r(d = 5\text{km}) = 5.4 \times 10^{-13} \text{ W} = -122.68 \text{ dBW or } -92.68 \text{ dBm}$$

57

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Q: Compute the diffraction loss for the three cases shown in Figure. Assume $\lambda = 1/3$ m, $d_1 = 1$ km, $d_2 = 1$ km, and (a) $h = 25$ m, (b) $h = 0$, (c) $h = -25$ m.

Compare your answers using values from Figure, as well as the approximate solution given by Equation. For each of these cases, identify the Fresnel zone within which the tip of the obstruction lies.

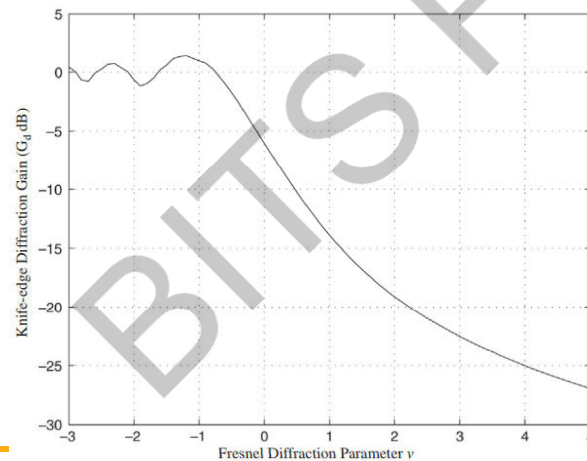
58

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$$v = h \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}}$$

$$G_d(\text{dB}) = 20 \log |F(v)|$$

Graphical representation of



Lee's approximate solution:

$$G_d(\text{dB}) = 0 \quad v \leq -1$$

$$G_d(\text{dB}) = 20 \log(0.5 - 0.62v) \quad -1 \leq v \leq 0$$

$$G_d(\text{dB}) = 20 \log(0.5 \exp(-0.95v)) \quad 0 \leq v \leq 1$$

$$G_d(\text{dB}) = 20 \log(0.4 - \sqrt{0.1184 - (0.38 - 0.1v)^2}) \quad 1 \leq v \leq 2.4$$

$$G_d(\text{dB}) = 20 \log\left(\frac{0.225}{v}\right) \quad v > 2.4$$

60

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Solution:

$$\lambda = 1/3, d_1 = 1 \text{ km}, d_2 = 1 \text{ km}.$$

(a) $h = 25 \text{ m}$,

$$v = h \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}} = 25 \sqrt{\frac{2(1000 + 1000)}{(1/3) \times 1000 \times 1000}} = 2.74$$

The diffraction loss is 22 dB.

The path length difference between the direct and diffraction rays is

$$\Delta \approx \frac{h^2}{2} \frac{(d_1 + d_2)}{d_1 d_2} = \frac{25^2 (1000 + 1000)}{2 \times 1000 \times 1000} = 0.625 \text{ m}$$

$$\Delta = n\lambda / 2 \Rightarrow n = \frac{2\Delta}{\lambda} = \frac{2 \times 0.625}{0.3333} = 3.75$$

61



(b) $h = 0 \text{ m}$, the Fresnel diffraction parameter $v = 0$.

From Figure 4.14 the diffraction loss is 6 dB.

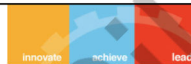
For this case $h = 0$ and $\Delta = 0$, the tip of the obstruction lies in the middle of the first Fresnel zone.

(c) $h = -25 \text{ m}$, the Fresnel diffraction parameter $v = -2.74$.

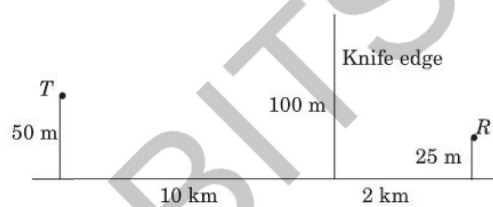
From Figure 4.14 the diffraction loss is 1 dB.

Using the numerical approximation in Equation(4.61a), the diffraction loss is equal to 0 dB.

62



Q: Given the following geometry, determine (a) the loss due to knife-edge diffraction, and (b) the height of the obstacle required to induce 6 dB diffraction loss. Assume $f = 900 \text{ MHz}$.



63



(a) The wavelength $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{900 \times 10^6} = \frac{1}{3} \text{ m}$

$$\beta = \tan^{-1}\left(\frac{75 - 25}{10000}\right) = 0.2865^\circ$$

$$\gamma = \tan^{-1}\left(\frac{75}{2000}\right) = 2.15^\circ$$

$$\alpha = \beta + \gamma = 2.434^\circ = 0.0424 \text{ rad}$$

$$v = 0.0424 \sqrt{\frac{2 \times 10000 \times 2000}{(1/3) \times (10000 + 2000)}} = 4.24$$

The diffraction loss is 25.5 dB.

(b) For 6 dB loss, $v = 0$.

$$\frac{h}{2000} = \frac{25}{12000}, \quad h = 4.16 \text{ m}.$$

64





Q: Four received power measurements were taken at distance of 100 m, 200 m, 1 km, and 3 km from a transmitter. These measured values are given in the following table. It is assumed that the path loss for these measurements follows the model in Equation(4.69a), where $d_0 = 100$ m:

- Find the minimum mean square error (MMSE) estimate for the path loss exponent, n ;
- Calculate the standard deviation about the mean value;
- Estimate the received power at $d = 2$ km using the resulting model;
- Predict the likelihood that the received signal level at 2 km will be greater than -60 dBm;
- Predict the percentage of area within a 2 km radius cell that receives signals greater than -60 dBm.

65



The sum of squared errors between the measured and estimated value is

$$J(n) = \sum_i^k (p_i - \hat{p}_i)^2$$

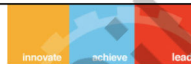
- (a) We find $\hat{p}_i = p_i(d_0) - 10n \log(d_i/100\text{m})$.

$$\hat{p}_1 = 0, \quad \hat{p}_2 = -3n, \quad \hat{p}_3 = -10n, \quad \hat{p}_4 = -14.77n.$$

$$J(n) = (0-0)^2 + (-20-(-3n))^2 + (-35-(-10n))^2 + (-70-(-14.77n))^2 \\ = 6525 - 2887.8n + 327.153n^2$$

$$\frac{dJ(n)}{dn} = 654.306n - 2887.8. \\ n = 4.$$

66



- (b) $\sigma^2 = J(n) / 4$ at $n = 4$.

$$J(n) = (0+0) + (-20+13.2)^2 + (-35+44)^2 + (-70+64.988)^2 \\ = 152.36.$$

$$\sigma^2 = 152.36 / 4 = 38.09 \text{ dB}^2, \quad \sigma = 6.17 \text{ dB}.$$

- (c) $d = 2$ km,

$$\hat{p}(d = 2\text{km}) = 0 - 10(4.4) \log(2000/100) = -57.24 \text{ dBm}.$$

$$(d) P_r[P_r(d) > -60 \text{ dBm}] = Q\left(\frac{\gamma - \overline{P_r(d)}}{\sigma}\right) = Q\left(\frac{-60 + 57.24}{6.17}\right) = 67.4\%$$

- (e) Equation (4.78) or Figure 4.18 may be used to determine that 88% of the cell area receives coverage above -60 dB.

67



HOME WORK 4.1

T1: Problems: 4.1, 4.2, 4.4, 4.5, 4.6

68





IMP Note to Self



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IMP Note to Self



IMP Note to Students

- It is important to know that just login to the session does not guarantee the attendance.
- Once you join the session, continue till the end to consider you as present in the class.
- IMPORTANTLY, you need to make the class more interactive by responding to Professors queries in the session.
- **Whenever Professor calls your number / name ,you need to respond, otherwise it will be considered as ABSENT**



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Modular Content Structure

- M1: Introduction to Wireless Systems and Modern Wireless Communication System.
- M2: The Cellular Concept- System Design Fundamentals
- M3: Cellular Network
- M4: Mobile Radio Propagation: Large scale path Loss_ Part 1
- M5: Mobile Radio Propagation: Large scale path Loss_ Part 2**
- M6: Mobile Radio Propagation: Large scale path Loss_ Part 3
- M7: Small Scale fading_ Part 1
- M8: Small Scale fading_ Part 1